An explicit polynomial of degree 29,984 in Smale's mean value problem

Abstract

We specify a Gaussian-rational polynomial of degree 29,984 whose normalized divided difference exceeds one at every critical point. Its normalization point is independent of the anchor of the analytic model. A complete interval calculation isolates all 29,983 critical points and encloses their normalized quotients. We give the finite definition, local root and value estimates, and outward-rounding arguments used by the verifier. A local comparison shows that the same polynomial has a critical quotient below one at its model anchor.

1 Result and scope

Theorem 1. *The exact polynomial $p \in \mathbb{Q}(i)[x]$ defined in Section 2, with the finite coefficient tables in* `polynomial.json`*, satisfies*

$$\deg p = 29\,984, \qquad p(0) = 0, \qquad p'(0) = 1,$$

and

$$\boxed{1.00000049595801 < \mathcal{S}(p,0) := \min_{p'(c)=0} \left| \frac{p(c)}{c} \right| < 1.00000049595804.} \tag{1}$$

All 29,983 critical points are simple. Exactly one critical point attains the minimum.

The finite specification and the rational containing disks suffice to reproduce the interval proof. The computation isolates and checks every critical point of this polynomial.

The model anchor a and the normalization base $\widehat{a}$ are different. The letter b denotes the rational approximation to an algebraic constant. Every numerical string, including terminating decimals and scientific notation, denotes an exact rational number. The specified phase has modulus one exactly.

2 Explicit finite polynomial

Set

$$m = 16, \quad N = 1874, \quad L = N - 2 = 1872, \qquad A = \frac{1061603}{250000}, \quad C = 1 + A^2, \quad S = A + 1/A,$$

$$q(z) = \frac{z^{16} + iA}{S}, \qquad v(z) = iq(z).$$

The exact amplitude, Cayley parameter, and phase are

$$\gamma = 0.61480744530374742407, \qquad t = 55.5578744900222005754, \qquad \omega = \frac{1+it}{1-it}.$$

Thus $\gamma > 0$ and $|\omega| = 1$ exactly. The normalization base is

$$\boxed{\widehat{a} = 1.040152313393317706 77 + 0.302102329270430736141\, i.}$$

The other Gaussian-rational parameters a, r, b, ξ_1, ξ_2 are printed in the appendix and supplied in the JSON specification.

Define

$$d_0 = C + \bar{r}^{16}(-A^2 + iA), \quad d_1 = iA - \bar{r}^{16}(1 + iA), \quad \alpha = -\frac{Cd_1}{iAd_0}.$$

For $1 \le k \le 15$, let $\nu_k = 1 - k/16$ and define

$$b_{k,0} = 1, \quad b_{k,j} = b_{k,j-1}\frac{\nu_k - j + 1}{j}, \qquad c_{k,0} = 1, \quad c_{k,j} = \alpha c_{k,j-1} + b_{k,j} \quad (1 \le j \le L).$$

Thus $b_{k,j} = \binom{\nu_k}{j}$. With

$$J(z) = -rC(1 + v(z)) + \bar{r}^{15}(1 + iA)z^{16},$$

define

$$\begin{aligned} h_L(z) = \frac{J(z)}{d_0}\sum_{j=0}^{L-1}(\alpha v(z))^j \\ + \frac{(1-|r|^2)C}{d_0}\sum_{k=1}^{15}\bar{r}^{k-1}b^k z^k \sum_{j=0}^{L} c_{k,j} v(z)^j. \end{aligned} \tag{2}$$

The first sum ends at $L-1$, the other sums at L. The b in this formula is the specified rational complex number, not an unspecified algebraic root.

For the corrections, set

$$K(z) = (z - \xi_1)^2(z - \xi_2)^2, \qquad \mu = (\xi_1 + \xi_2)/2,$$

$$R_0(z) = \sum_{\ell=0}^{63}\sum_{k=0}^{15} d_{\ell k}\left(\frac{z}{6/5}\right)^k q(z)^\ell, \qquad T(z) = \sum_{j=0}^{3}\eta_j(z-\mu)^j,$$

$$\Delta_0(z) = (z-a)^2[K(z)R_0(z) + T(z)], \qquad X(z) = (z-a)^2\sum_{\ell=0}^{127}\sum_{k=0}^{15} x_{\ell k}\left(\frac{z}{6/5}\right)^k q(z)^\ell.$$

The arrays d, η, x are respectively `correction_coefficients`, `free_coefficients`, and `extra_correction_coefficients` in the specification. The retained Hermite coefficient is exactly zero. The high-degree tail is

$$E(z) = \sum_{\ell=0}^{7}\sum_{k=0}^{15} t_{\ell k}\left(\frac{z}{6/5}\right)^k q(z)^\ell, \qquad H(z) = \omega[\gamma q(z)^{1874} + q(z)^{1866}E(z)],$$

where $t_{\ell k}$ is the table `tail_coefficients`, not the scalar Cayley parameter t. Define

$$P(z) = (z-a)\frac{h_L(z)}{h_L(a)} + \Delta_0(z) + X(z) + H(z) \tag{3}$$

and finally

$$\boxed{p(x) = \frac{P(\widehat{a}+x) - P(\widehat{a})}{P'(\widehat{a})}.} \tag{4}$$

Both corrections vanish to order two at a, whereas normalization is at $\widehat{a}$. The evaluator uses the full finite values $P(\widehat{a})$ and $P'(\widehat{a})$ in (4), and encloses the latter away from zero. This gives $p(0) = 0$ and $p'(0) = 1$ exactly.

The degree bounds are

term	degree, at most
h_L	29 967
$(z-a)h_L$	29 968
Δ_0	1029
X	2049
$q^{1866}E$	29 983.

Only $\omega\gamma q^{1874}$ contributes in degree 29,984. Its coefficient is nonzero, so translation and the nonzero normalization in (4) preserve degree exactly.

3 An independent base point

The anchor a defines the analytic model, while the critical-point functional can be evaluated at any base point b_0 with $P'(b_0) \neq 0$. For a fixed polynomial and fixed critical point c, define

$$Q_c(b_0) = \frac{P(c) - P(b_0)}{(c - b_0)P'(b_0)}.$$

Where the denominators and numerator are nonzero,

$$\frac{d}{db_0} \log Q_c(b_0) = -\frac{P'(b_0)}{P(c) - P(b_0)} + \frac{1}{c - b_0} - \frac{P''(b_0)}{P'(b_0)}. \tag{5}$$

Changing b_0 leaves every critical point fixed. The base point therefore supplies two real parameters without changing the degree. The normalization in (4) follows after the polynomial and base point are chosen.

The local computation in `base_comparison.log` isolates a root of P' in containing disk 27158 for which

$$0.99982426177940 < \left| \frac{P(c) - P(a)}{(c - a)P'(a)} \right| < 0.99982426177944 < 1.$$

Thus the independent base point is essential for this example: at the model anchor a, the same polynomial has a critical quotient below one.

4 Complete critical-point coverage

Computations use the P-coordinate. A critical point c corresponds to $c - \widehat{a}$ for p, and the required quotient is

$$Q(c) = \left| \frac{P(c) - P(\widehat{a})}{(c - \widehat{a})P'(\widehat{a})} \right|. \tag{6}$$

The file `critical_disks.txt` supplies exactly 29 983 rational centers and positive rational radii. These are *containing disks*; the accepted root disks may be smaller and may have recentered dyadic centers.

Lemma 1 (Local linear comparison)**.** *Let G, g be holomorphic on a neighborhood of $D = \{z : |z - z_0| \leq \rho\}$. Suppose $E \geq \sup_D |G - g|$ and $V \geq \sup_D |g'(z) - g'(z_0)|$. If*

$$|g(z_0)| + \rho V + E < \rho |g'(z_0)|, \tag{7}$$

then G has exactly one zero in the open disk, counted with multiplicity.

Proof. On the boundary, integration along the radial segment gives

$$|G(z) - g'(z_0)(z - z_0)| \le E + |g(z_0)| + \rho V < |g'(z_0)(z - z_0)|.$$

Rouché's theorem compares G with a nonzero linear polynomial, which has one zero inside [1]. In particular, the unique zero is simple. □

The first local test applies this lemma with $G = P'$ and with g the derivative of the analytic comparison function defined below. When its approximation error forces a needlessly large disk, the code uses finite-polynomial values at the center instead.

Lemma 2 (Finite-center comparison)**.** *Let $e = P' - g$ be holomorphic on a neighborhood of the disk of radius $R > \rho$ about z_0, and suppose $|e| \le E_R$ there. If $V \ge \sup_{|z-z_0|\le\rho} |g'(z) - g'(z_0)|$, then*

$$V_P = V + \frac{2\rho E_R}{(R-\rho)^2} \;\ge\; \sup_{|z-z_0|\le\rho} |P''(z) - P''(z_0)|.$$

Consequently,

$$|P'(z_0)| + \rho V_P < \rho |P''(z_0)| \tag{8}$$

proves exactly one simple critical point in the disk.

Proof. Cauchy's derivative estimate yields $|e''(z)| \le 2E_R/(R-\rho)^2$ on the smaller disk. Integrate along a radius to bound $e'(z) - e'(z_0)$, add the bound for g', and apply the same linear comparison to P'. □

Every containing-disk pair that is not already separated by its real-coordinate gap is checked using a rigorous lower bound for center distance and an upper bound for the sum of radii. The real-coordinate sweep uses the largest radius, so omitted pairs are provably disjoint. Every accepted root disk is checked to lie inside its corresponding containing disk. Finite Newton steps merely propose centers: only the subsequent strict inequalities certify a root.

There are 29 983 disjoint accepted disks, each containing one root with multiplicity, and $\deg P' = 29\,983$. The fundamental theorem of algebra therefore exhausts the critical set and proves that every critical point is simple. The sixteen work partitions are index classes modulo sixteen.

5 Analytic comparison at the fixed model anchor

Set $D = \Delta_0 + X$ and introduce, solely for the proof,

$$\beta = \left(\frac{1+iA}{C}\right)^{1/16}, \quad U = 1 + iq, \quad W = \beta z U^{-1/16}, \quad B(w) = \frac{w - r}{1 - \bar{r}w},$$

$$B_a = B(W(a)), \qquad G(z) = \frac{B(W(z))}{B_a}, \qquad P_*(z) = (z-a)G(z) + D(z) + H(z), \qquad g = P_*'.$$

Fractional powers have their principal values. In every comparison box, the code excludes the branch cut and all relevant poles. The segment-denominator check in Section 6 permits these local comparisons also when $|q| \ge 1$. The disjoint-disk degree count supplies complete critical-point coverage. Direct differentiation gives

$$W' = \frac{\beta}{C} U^{-1-1/16}, \quad G' = \frac{(1-|r|^2)W'}{(1-\bar{r}W)^2 B_a}, \quad G'' = G'\left[-\frac{17iAz^{15}}{1+iAz^{16}} + \frac{2\bar{r}W'}{1-\bar{r}W}\right].$$

The exact rational identity $W^{16} = (1 + iA)z^{16}/(1 + iAz^{16})$ gives

$$B(W) = \frac{J}{d_0(1 - \alpha v)} + \frac{(1 - |r|^2)C}{d_0} \sum_{k=1}^{15} \bar{r}^{k-1} \beta^k z^k \frac{(1 + v)^{\nu_k}}{1 - \alpha v}. \tag{9}$$

This is obtained by multiplying $B(W)$ by $\sum_{j=0}^{15} (\bar{r}W)^j$. It is finite algebra, not a convergence assertion about that geometric series.

5.1 The algebraic constant is rigorously isolated

Let $\epsilon = 2 \cdot 10^{-60}$ and $T_\beta(z) = z^{16} - (1 + iA)/C$. Using exact GMP rational arithmetic, the initializer verifies

$$|\Re T_\beta(b)| + |\Im T_\beta(b)| + \sum_{j=2}^{16} \binom{16}{j} (|\Re b| + |\Im b|)^{16-j} \epsilon^j < 16(\Re b)^{15} \epsilon.$$

Rouché's theorem therefore places one root within ϵ of b. The verified inequalities $\Re b > \epsilon$, $\Im b > \epsilon$, and $10(\Im b + \epsilon) < \Re b - \epsilon$ select the principal root. In particular,

$$|b^k - \beta^k| \le a_k := k\epsilon(|\beta| + \epsilon)^{k-1}.$$

The polynomial itself still uses $b \in \mathbb{Q}(i)$ throughout.

6 Finite binomial remainders beyond the Taylor circle

For $0 < \nu < 1$, define

$$T_L^\nu(v) = (1 + v)^\nu - \sum_{j=0}^{L} \binom{\nu}{j} v^j.$$

Lemma 3. *For $v \notin (-\infty, -1]$,*

$$T_L^\nu(v) = \frac{(-1)^L \sin(\pi\nu)}{\pi} v^{L+1} \int_0^1 \frac{t^{L-\nu}(1 - t)^\nu}{1 + tv}\, dt. \tag{10}$$

Set $u = |v|$, $\sigma(v) = \min_{0 \le t \le 1} |1 + tv|$, and $b_L = |\binom{\nu}{L+1}|$. Then

$$|T_L^\nu(v)| \le b_L u^{L+1}/\sigma, \tag{11}$$

$$|(T_L^\nu)'(v)| \le b_L \left(\frac{(L + 1)u^L}{\sigma} + \frac{L + 1 - \nu}{L + 2} \frac{u^{L+1}}{\sigma^2} \right). \tag{12}$$

These formulas do not require $|v| < 1$.

Proof. For $|v| < 1$, expand $(1 + tv)^{-1}$ and apply Euler's beta integral [2]. Its coefficients are exactly the binomial tail coefficients, including their signs. Both sides of (10) are holomorphic on the slit plane, so analytic continuation gives the identity there. The integral's total mass, after multiplication by $\sin(\pi\nu)/\pi$, is b_L. Its normalized first moment in t is $(L + 1 - \nu)/(L + 2)$. Taking absolute values, and then differentiating, proves (11)–(12). □

For a complex box with point midpoint c and enclosing disk radius r_0, $\sigma(v) \ge \sigma(c) - r_0$. The value $\sigma(c)$ is the distance from zero to the segment from 1 to $1 + c$: it is 1, $|1 + c|$, or $|\Im c|/|c|$, according as the minimizing projection lies at an endpoint or in the interior. When interval tests

cannot determine the case uniquely, the smaller candidate bound is used. A strict positive lower bound both bounds the integral and excludes the binomial branch cut on the entire box.

The recurrence for $c_{k,j}$ implies the finite identity

$$\mathcal{R}_{k,L}(v) := \frac{(1+v)^{\nu_k}}{1-\alpha v} - \sum_{j=0}^{L} c_{k,j} v^j = \frac{T_L^{\nu_k}(v) + \alpha c_{k,L} v^{L+1}}{1-\alpha v}.$$

With $d = |1-\alpha v|$, take

$$T_{k,0} = \frac{b_L u^{L+1}/\sigma + |\alpha c_{k,L}| u^{L+1}}{d},$$

$$T_{k,1} = \frac{b_L[(L+1)u^L/\sigma + (L+1-\nu_k)u^{L+1}/((L+2)\sigma^2)] + (L+1)|\alpha c_{k,L}|u^L}{d} + \frac{|\alpha| T_{k,0}}{d}.$$

They bound $|\mathcal{R}_{k,L}|$ and $|\mathcal{R}'_{k,L}|$ respectively. All denominator lower bounds are checked to be positive.

6.1 Combining all analytic errors

Set $B_0 = |1+v|$, $t = |\alpha| u$, and

$$w_k = \frac{(1-|r|^2)C|r|^{k-1}}{|d_0|}, \qquad H_k = B_0^{\nu_k}/d, \qquad H_{k,1} = \nu_k B_0^{\nu_k - 1}/d + |\alpha| B_0^{\nu_k}/d^2.$$

The finite geometric identity and (9) give

$$\begin{aligned} e_h &= \frac{|J| t^L}{|d_0| d} + \sum_{k=1}^{15} w_k |z|^k (|b|^k T_{k,0} + a_k H_k), \\ e_{h'} &= \frac{|J'| t^L}{|d_0| d} + \frac{|J||\alpha||v'|}{|d_0|}\left(\frac{L t^{L-1}}{d} + \frac{t^L}{d^2}\right) \\ &\quad + \sum_{k=1}^{15} w_k \left[k|z|^{k-1}(|b|^k T_{k,0} + a_k H_k) + |z|^k |v'|(|b|^k T_{k,1} + a_k H_{k,1})\right]. \end{aligned}$$

These bound $|h_L - B(W)|$ and the difference of their derivatives. The initializer proves $|h_L(a) - B_a| < \epsilon_a = 10^{-50}$ and $h_a^- := |B_a| - \epsilon_a > 0$. Hence, for $G_L = h_L / h_L(a)$,

$$|G_L - G| \le e_g := \frac{e_h + |G|\epsilon_a}{h_a^-}, \qquad |G'_L - G'| \le e_{g'} := \frac{e_{h'} + |G'|\epsilon_a}{h_a^-}.$$

It follows that $|P' - g| \le e_g + |z-a| e_{g'}$. The polynomial corrections and high-degree tail cancel exactly from this difference. Every expression is evaluated on the entire current box, not on a set of samples.

7 Finite-polynomial evaluation at a center

The code has two implementations of h_L, h'_L, h''_L. Direct finite Horner evaluation propagates values and two derivatives through the rational coefficient recurrence. The faster implementation subtracts a rigorously enclosed *finite* remainder from $(1+v)^\nu$.

Normalize the beta weight in (10) to a probability measure and define $\mu_j = \mathbb{E}[(1-t)^j] = (\nu+1)_j/(L+2)_j$. A finite geometric identity implies, for any integer $K \ge 0$,

$$\mathbb{E}\frac{1}{1+tv} = \sum_{j=0}^{K-1} \frac{\mu_j v^j}{(1+v)^{j+1}} + \varepsilon_K, \qquad |\varepsilon_K| \le \frac{1}{\sigma(v)} \left(\frac{|v|}{|1+v|}\right)^K \mu_K.$$

For $s = 1 - t$, the finite geometric remainder is $[sv/(1+v)]^K/(1+tv)$, which proves the bound directly.

The implementation selects the best bound among $0 \le K \le 32$ on a slightly larger box. If E bounds the error in T_L^ν there, and a disk of radius R about every center-box point lies in that larger box, Cauchy's estimate bounds the errors in its first two derivatives by E/R and $2E/R^2$. The branch and denominator checks are repeated on the larger box.

For clarity, let $S_K(v)$ denote the finite moment sum and write $A_{k,L}(v) = \sum_{j=0}^{L} c_{k,j} v^j$. With $e_k = \alpha c_{k,L}$,

$$A_{k,L} = \frac{(1+v)^{\nu_k} - T_L^{\nu_k} - e_k v^{L+1}}{1 - \alpha v},$$

$$A'_{k,L} = \frac{\nu_k (1+v)^{\nu_k - 1} - (T_L^{\nu_k})' - (L+1) e_k v^L + \alpha A_{k,L}}{1 - \alpha v},$$

$$A''_{k,L} = \frac{\nu_k (\nu_k - 1)(1+v)^{\nu_k - 2} - (T_L^{\nu_k})'' - L(L+1) e_k v^{L-1} + 2\alpha A'_{k,L}}{1 - \alpha v}.$$

These identities account for every finite-tail term. If the moment enclosure is too wide or a local analytic check is inconclusive, the code falls back to the direct finite recurrence. Neither implementation substitutes the limiting analytic function for the finite polynomial.

The polynomial corrections are evaluated in their factored, compressed bases, including the additional 128 blocks of X. The tail uses the exact product rule for $H = \omega q^{1866}(\gamma q^8 + E)$. Consequently the evaluated finite jets are

$$\begin{aligned} P &= (z-a) h_L / h_L(a) + D + H, \\ P' &= [h_L + (z-a) h'_L] / h_L(a) + D' + H', \\ P'' &= [2h'_L + (z-a) h''_L] / h_L(a) + D'' + H''. \end{aligned}$$

8 Quotient enclosures at the independent base

This section gives the new formulas used by the verifier. Set

$$V = P(\widehat{a}), \qquad D_\star = P'(\widehat{a}), \qquad T_*(z) = P_*(z), \qquad g = T'_*,$$

where $V, D_\star$ are directly enclosed by the finite polynomial evaluator. The initializer proves $|D_\star| > 0$. Define

$$E_0 = |z-a| e_g, \qquad E_1 = e_g + |z-a| e_{g'}.$$

The preceding error bounds give $|P - T_*| \le E_0$ and $|P' - g| \le E_1$. For a disk $D = \{|z - z_0| \le \rho\}$ avoiding $\widehat{a}$, define

$$A_*(z) = \frac{T_*(z) - V}{z - \widehat{a}}, \qquad \widetilde{Q}(z) = \frac{T_*(z) - V}{(z - \widehat{a}) D_\star}.$$

Then

$$\widetilde{Q}'(z) = \frac{g(z) - A_*(z)}{(z - \widehat{a}) D_\star}, \qquad \left| \frac{P(z) - V}{(z - \widehat{a}) D_\star} - \widetilde{Q}(z) \right| \le \frac{E_0}{|z - \widehat{a}||D_\star|}.$$

Consequently the model-center enclosure has center modulus $|\widetilde{Q}(z_0)|$ and error radius

$$\rho \sup_D |\widetilde{Q}'| + \sup_D \frac{E_0}{|z - \widehat{a}||D_\star|}. \tag{13}$$

Subtract this radius for a lower bound and add it for an upper bound. All suprema and center values are evaluated by outward intervals.

For the finite-center test, use the directly evaluated modulus

$$q_0 = \left| \frac{P(z_0) - V}{(z_0 - \widehat{a})D_\star} \right|.$$

Since

$$\left| P'(z) - \frac{P(z) - V}{z - \widehat{a}} \right| \le |g(z) - A_*(z)| + E_1 + \frac{E_0}{|z - \widehat{a}|},$$

a whole-disk variation bound is

$$R_Q = \rho \sup_D \frac{|g - A_*| + E_1 + E_0/|z - \widehat{a}|}{|z - \widehat{a}||D_\star|}. \tag{14}$$

The interval $[q_0 - R_Q, q_0 + R_Q]$ encloses the critical quotient in the disk. These formulas follow by differentiation and integration along a radius; they do not assume that the center is an exact critical point. The interval divisions prove that every accepted disk avoids $\widehat{a}$. Uncertainty in V and $D_\star$ is retained in every operation.

The exhaustive run alone proves

$$1.00000000407528 < \mathcal{S}(p, 0) < 1.00000052222008.$$

The refinement retests 316 potentially limiting containing disks. Each contains exactly one root by the exhaustive stage, so its refined and original quotient intervals can be intersected. All other disks retain their original bounds. The resulting global enclosure is

$$1.00000049595801828345 < \mathcal{S}(p, 0) < 1.00000049595803927317.$$

The minimizing disk has index 28891. For every other critical point,

$$Q(c) > 1.0000004992776471.$$

This lower bound exceeds the upper bound for the minimizing disk, so the minimizer is unique.

9 Interval arithmetic and numerical trust assumptions

The backend encloses basic real operations by rounding the computed `long double` endpoint one representable number outward. It runs in IEEE round-to-nearest mode with binary arithmetic and at least 64 significand bits. Rational constants are first enclosed with GMP/MPFR conversion. Nonfinite intervals, zero-containing divisors, and unsupported formats are rejected. Transcendental endpoints are computed by MPFR at 192 bits with rounding toward the appropriate infinity [3]. Compiler flags disable fast-math reassociation and contraction of multiply-add expressions.

9.1 Floating-center complex disks

A complex enclosure is $D(c, r)$, with a floating-point center and an outward radius. Before center roundoff is added, the exact rules are

$$D(c, r) + D(d, s) \subset D(c + d, r + s), \qquad D(c, r)D(d, s) \subset D(cd, |c|s + |d|r + rs).$$

Let $u = 2^{-p}$ for significand precision p, and let τ be the least positive normal number of the hardware type. It dominates the subnormal rounding floor. If $M_c \ge |c|$ and $M_d \ge |d|$, the implementation adds

$$4u(M_c + M_d) + 8\tau \quad \text{for complex addition,}$$

$$8uM_cM_d + 32\tau \quad \text{for complex multiplication.}$$

These are conservative upper bounds for the Euclidean error of the computed center. To verify them, use the scalar round-to-nearest bound $|\,\mathrm{fl}(x) - x| \le u|x| + \eta/2$, where η is the least positive subnormal. The two additions have total absolute error at most $\sqrt{2}u(M_c + M_d) + \eta$. For a complex product, four products and two additions give total absolute error at most $(4u + 2u^2)M_cM_d + (3 + 2u)\eta$. The displayed allowances dominate both for $p \ge 64$. A similar scalar-multiplication bound includes the radius of any interval scalar. Magnitudes may be replaced by their larger ℓ^1 bounds; all radius computations remain outward interval operations.

Horner propagation carries the value and its first two derivatives. For example, $(Uq + w)'' = U''q + 2U'q' + Uq'' + w''$. Fixed interval coefficients are converted to enclosing disks; their conversion error is added to the radius. These rules apply on a complete input box as well as at a center. They avoid repeated rectangular wrapping but do not change any polynomial coefficient. The exact-rational regression program performs 64,800 image-containment tests, including subnormal and extreme-exponent inputs. It supplements this error argument rather than replacing it.

High powers are evaluated by rotating the input through a quarter turn into a verified right half-plane and using interval modulus and argument. The rotation is undone algebraically. All trigonometric argument ranges used by this certificate are bounded; the endpoint/extremum enumeration in the interval backend includes every possible extremum on those ranges. Wide intervals are enclosed by $[-1, 1]$ instead.

The calculation relies on these enclosure arguments, the supplied source, GMP/MPFR, the compiler with the stated flags, and the hardware semantics specified above.

10 Complete verification, audits, and reproduction

The exhaustive wrapper requires all sixteen work partitions to pass and checks that the recorded critical-point indices are exactly $0, \ldots, 29982$, each once. The partitions are index classes modulo sixteen; they are not geometric sheets used to infer root coverage. The containing disks are rational data, and their disjointness and the root counts do not depend on the numerical search.

`make_parameters.py` checks the exact coefficient tables and both base-point coordinates against the native rational input serialization. `run_verification.py` compiles the verifier and runs all partitions. `refine_certificate.py` binds its sharper calculations to the hashes and complete coverage of the exhaustive stage. `audit_certificate.py` checks degree bounds, data identity, coverage, and the combined intervals. `audit_exact_disks.py` independently checks rational containing-disk disjointness, accepted-disk containment, and every recorded strict root-count comparison. `audit_normalization.py` additionally checks the new base parameters and the nonzero derivative enclosures in all partition logs.

The exact-rational audit expands printed endpoints outward. The base summary widens endpoints by 10^{-15}, while the refinement and coverage audit use $(1 + |\text{endpoint}|)10^{-18}$. Published endpoints are rounded outward again. An integrity audit checks recorded calculations; it does not replace the native analytic enclosure computation.

From the repository directory, run

```
python3 certify.py --jobs 4
```

This runs exact input checks, local arithmetic and anchor comparisons, exhaustive verification, refinement, and the coverage and normalization checks. It requires Python 3.10 or later, a C++17 compiler, GMP with C++ headers, and MPFR on the supported binary `long double` platform. The README gives the build commands and arithmetic requirements. To check the supplied completed calculation without rerunning its interval operations, use `check_stored_certificate.py`.

A Exact seed parameters and coefficient tables

Every terminating decimal in this appendix is exact. The coefficient arrays are in the accompanying JSON specification.

$$\Re a = 1.040809364274614813715752515609076641243 7.$$

$$\Im a = 0.304190216501749670526226428656847209994 6.$$

$$\Re r = 0.986946158825205068813747279496424837686 7.$$

$$\Im r = -0.043244856253068185952044355025633762776 1.$$

$$\Re b = 0.908850253183765306046968000840344747056738618388026628837241.$$

$$\Im b = 0.076267057870218869943792292811043863508717811372210553566224.$$

$$\Re \xi_1 = 0.949453054880863382597892820880833672746761428093 37.$$

$$\Im \xi_1 = 0.012956094504147205529697045250153901100373436381 89.$$

$$\Re \xi_2 = 0.948450124975315957237918297288134097817186535119 44.$$

$$\Im \xi_2 = 0.013274296965271082358322583775608905580829913127 89.$$

The normalization base $\widehat{a}$, amplitude γ, and phase parameter are given in Section 2. The JSON contains all 1,024 coefficients of R_0, four cubic-remainder coefficients, 2,048 coefficients of $X/(z-a)^2$, and 128 coefficients of E. These tables and the finite recurrence define the polynomial without a monomial expansion.

References

[1] NIST Digital Library of Mathematical Functions, §1.10(iv), Rouché's theorem, and §1.11(i), polynomial zeros. `https://dlmf.nist.gov/1.10` and `https://dlmf.nist.gov/1.11`.

[2] NIST Digital Library of Mathematical Functions, §5.12, Euler's beta integral, equation 5.12.1. `https://dlmf.nist.gov/5.12`.

[3] GNU MPFR Manual, rounding modes and correct rounding. `https://www.mpfr.org/mpfr-current/mpfr.html`.